25901794 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCSTA100 - INTRODUCTION TO STATISTICAL MODELLING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **7** questions

Each question carries **2** marks

1. Given the pdf of X as f(x) = 2x ; 0 < x < 1 and 0 elsewhere, find pdf of Y = 3X + 1. [U] / [CO1]
2. Define Harmonic mean in terms of expectation [U] / [CO2]
3. Give any two merits of normal distribution. [U] / [CO3]
4. Mention the two properties of *pdf* of a random variable. [U] / [CO1]
5. Mention any two properties of Moment Generating Function. [U] / [CO2]
6. If a Poisson distribution has a mean of 4, what is the probability of observing exactly 2 events? [U] / [CO3]
7. Define a discrete random variable. Give an example. [U] / [CO1]
8. The time required to finish a task is normally distributed with mean 45 minutes and standard deviation 5 minutes. What is the probability that a randomly chosen person finishes in less than 40 minutes? [A] / [CO3]

9. Mention any two properties of geometric distribution. [A] / [CO3]

10. Define Poisson distribution. [K] / [CO3]

[2x7 = 14]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **6** marks

11. Define distribution function of a continuous random variable. Mention any three properties of it. [K] / [CO1]

12. *Given the pdf of X as $f(x) = 30\,x^4\,(1-x)$; $0<x<1$ and 0 elsewhere. Find E(X).* [A] / [CO2]

13. The mean and variance of a binomial distribution are 4 and 3 respectively. Write down its pmf. [A] / [CO3]

14. If X is Bernoulli variate taking values 0 and 1 with probabilities 0.4 and 0.6 respectively, find its mean and variance. [U] / [CO3]

15. Mention the properties of continuous uniform distribution. [A] / [CO3]

16. Mention the properties of exponential distribution. [An] / [CO3]

[6x4 = 24]

Part C

Essay Questions

Answer any **1** questions

Each question carries **12** marks

17. Given the *pmf* of a random variable as $f(x) = kx^2$; x = 1, 2, 5 and 0 elsewhere. Find *k*. Also find P (X > 1.7), P (1 < X < 3), P (X takes an even number), P ($X \leq 3.4$). [U] / [CO1]

18. A basket contains 20 bad oranges and 80 good oranges. Three are drawn at random from this basket. Find the probability that (a) exactly 2 are good (b) at least 2 are good (c) at most two are good. [A] / [CO3]

[12x1 = 12]